Araz Mehrabani

Mechanical Engineer | Wind Energy · Loads & Simulation · CAE

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Portfolio: https://arazmehrabani.github.io/

WORK EXPERIENCE

Development Engineer **01/2022–02/2023**
Paydar | Renewable Energy Systems — Tehran

- Supported small wind turbine projects: load calculations with **OpenFAST**, structural stress analyses in **ANSYS**, and site assessment with **QGIS**.
- Designed an electromechanical cleaning system for PV modules in SOLID-WORKS through manufacturing, assembly, and installation; design optimizations reduced scrap and raw-material use by approx. 10%.

Mechanical Engineer **03/2017–12/2021**
TehranRigzar | Mining and Mineral Processing Equipment — Tehran

- Co-developed a high-frequency vibrating screen; replaced the crankshaft drive concept with vibration motors to improve maintainability and reliability.
- Performed FEA-based vibration and resonance analyses in **ANSYS** to evaluate and optimize dynamic behavior; mechanical design in **CATIA V5**.

ADDITIONAL EXPERIENCE

Student Research Assistant — Hakim Sabzevari University — Underwater Robotics & Simulation **12/2014–12/2016**

Junior Project Engineer — Ario Fidar Maham Co. — Tehran **11/2012–05/2013**

EDUCATION

M.Sc. Wind Energy Engineering **03/2023–Present**
Flensburg University of Applied Sciences | current grade: 1.9 (German scale)

M.Sc. Mechanical Engineering **09/2014–01/2017**
Hakim Sabzevari University | Control Engineering and Robotics

B.Sc. Mechanical Engineering **09/2008–06/2013**
Islamic Azad University | Structural Dynamics

SELECTED PROJECTS

20 MW Offshore Wind Turbine
OpenFAST simulations and load/fatigue analysis according to IEC 61400-1; post-processing with Python and MATLAB.

Autonomous Underwater Welding Robot
Motion control in MATLAB/Simulink, modeling and simulation as part of the M.Sc. thesis; four peer-reviewed publications.

TECHNICAL SKILLS

Wind Energy & Loads
OpenFAST · IEC 61400-1 · DNV-ST-0437 · load and fatigue analysis

Site Assessment & Planning
windPRO · QGIS

CAE / Structural Dynamics
ANSYS Workbench · FEA / FEM · vibration and resonance analysis

Programming & Data Analysis
Python · MATLAB/Simulink · Power BI

CAD & Design
CATIA V5 · SOLIDWORKS · Siemens NX · Inventor

LANGUAGES

German	B1
English	C1 / IELTS 7.0
Persian	Native

TRAINING

IEEE ANSYS Training
HarvardX Python Programming
SOLIDWORKS Training
CATIA V5 Training

FOCUS AREAS

Wind Energy & Loads
Aeroelastic Simulation
Site Assessment
FEA / Structural Dynamics

More projects and work samples: arazmehrabani.github.io